

ANANTA AI : NL → SQL Query Builder for SQLite

Presented by

Savitha S Sivaselvi S Divyadharshini R
Ganikarika M

Project Overview

- SQL-01 is a specialized Natural Language (NL) to SQL Query Builder designed to facilitate data access for non-technical users.
- The system functions as a schema-grounded text-to-SQL agent that translates user intent expressed in plain English into functional SQL queries.
- It is built as a Streamlit application that serves as an interactive interface between the user and the SQLite engine.

Problem Statement

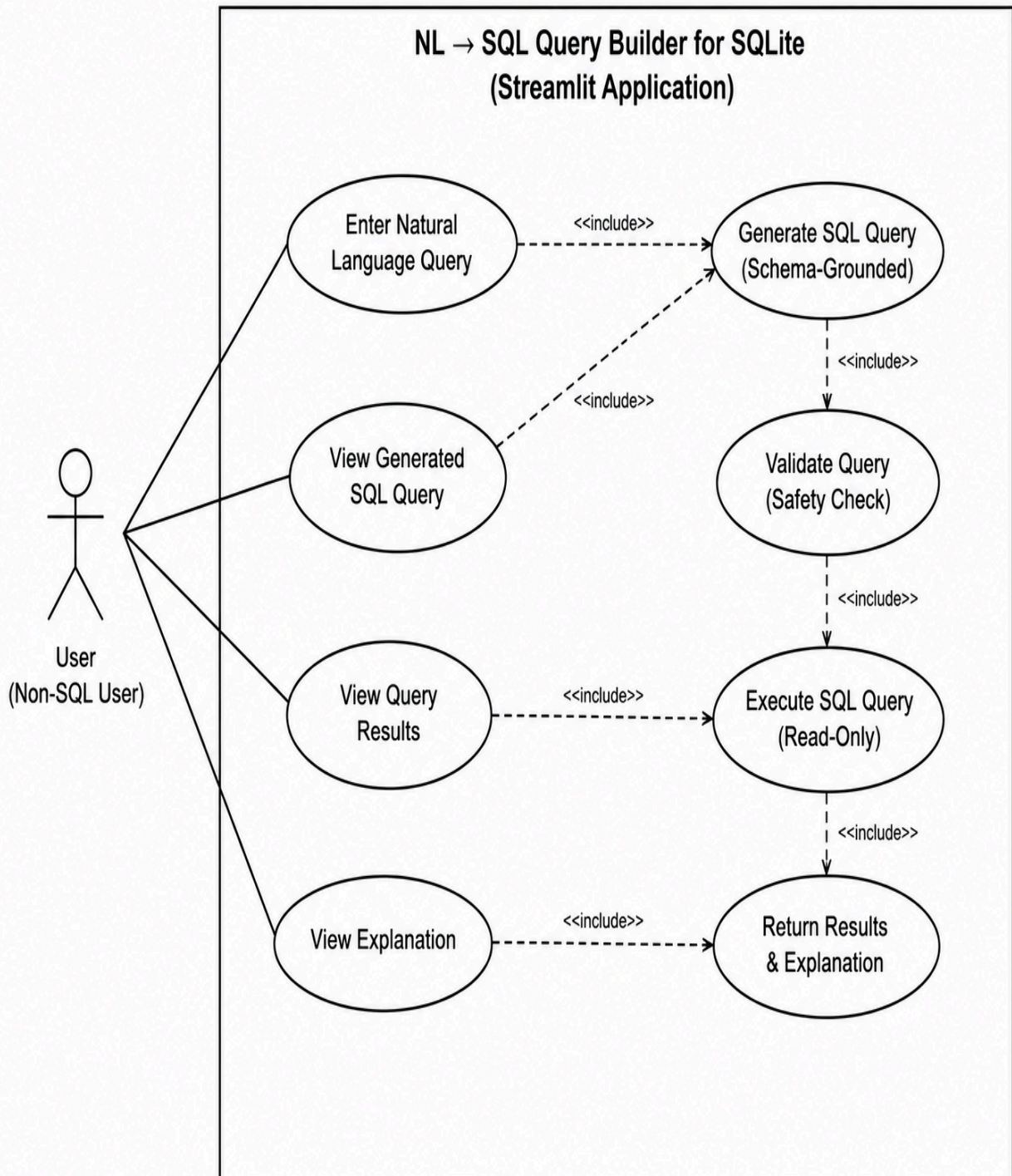
- Many users have valuable data locked within SQLite databases but lack the technical expertise required to perform complex operations, such as multi-table joins.
- Standard SQL syntax acts as a barrier for non-technical stakeholders who need to perform data analysis.
- There is a critical need for a secure, accessible bridge that allows users to explore data autonomously without needing to manually write structured code.

System Architecture

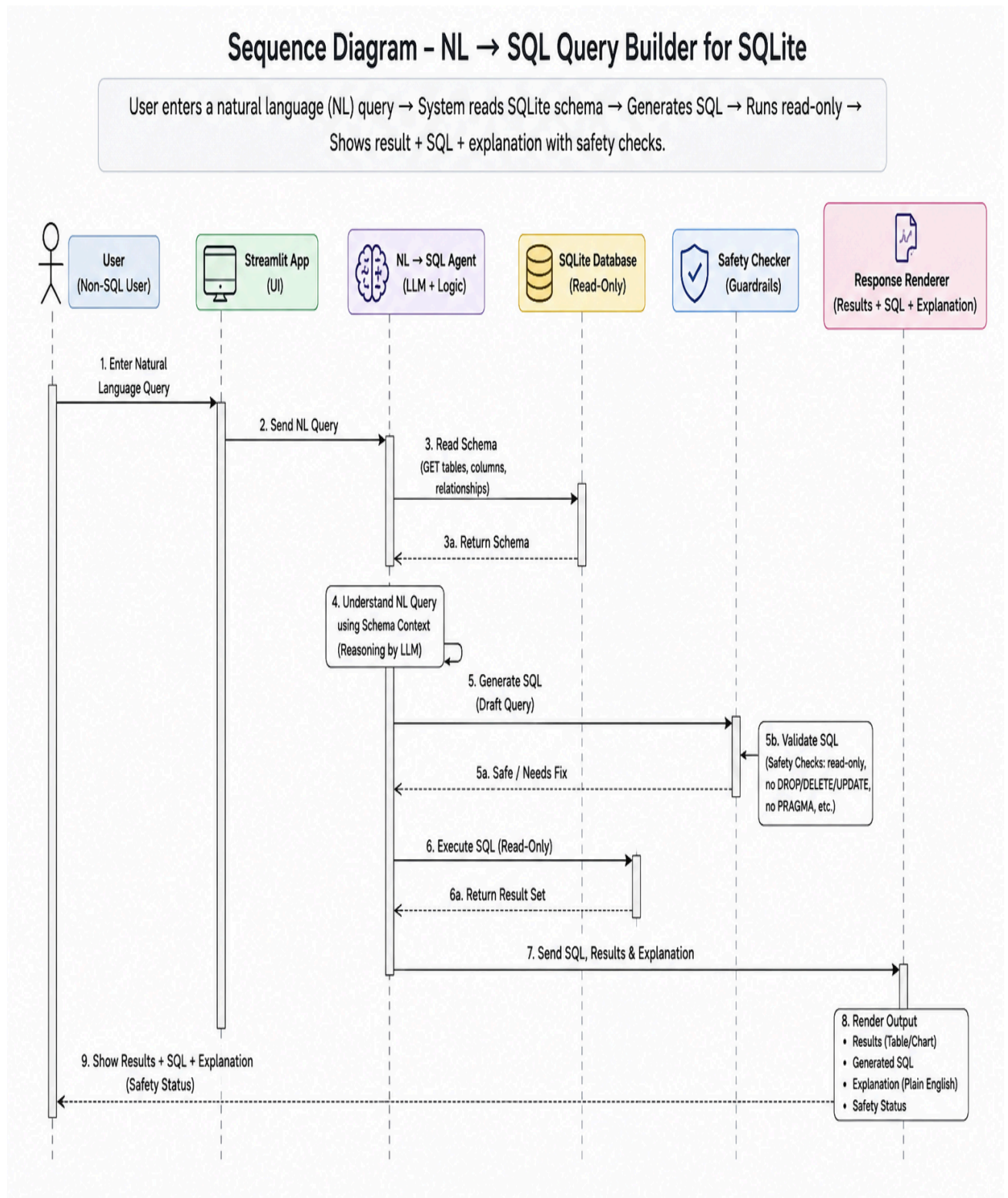
- The system utilizes a modular architecture divided into five distinct layers.
- The frontend is powered by Streamlit to facilitate state-managed interactions.
- The backend operates as an asynchronous SQLite controller to maintain thread safety.
- Infrastructure components include:

- **Natural Language Processing (NLP) Engine:** Responsible for converting plain text queries into functional SQL.
- **Schema Introspector:** Dynamically updates system context by reading the database file structure.
- **Validator/Guardrail:** A dedicated "Safety Layer" that sanitizes outputs before execution.
- **SQLite Execution Engine:** Functions as a restricted, read-only terminal for query execution.
- **Query Orchestrator (Controller Layer):**
Acts as the central coordinator that manages the flow between NLP Engine, Schema Introspector, Validator, and Execution Engine. It ensures that each module is executed in sequence, handles intermediate data transfer, and maintains overall system workflow consistency.

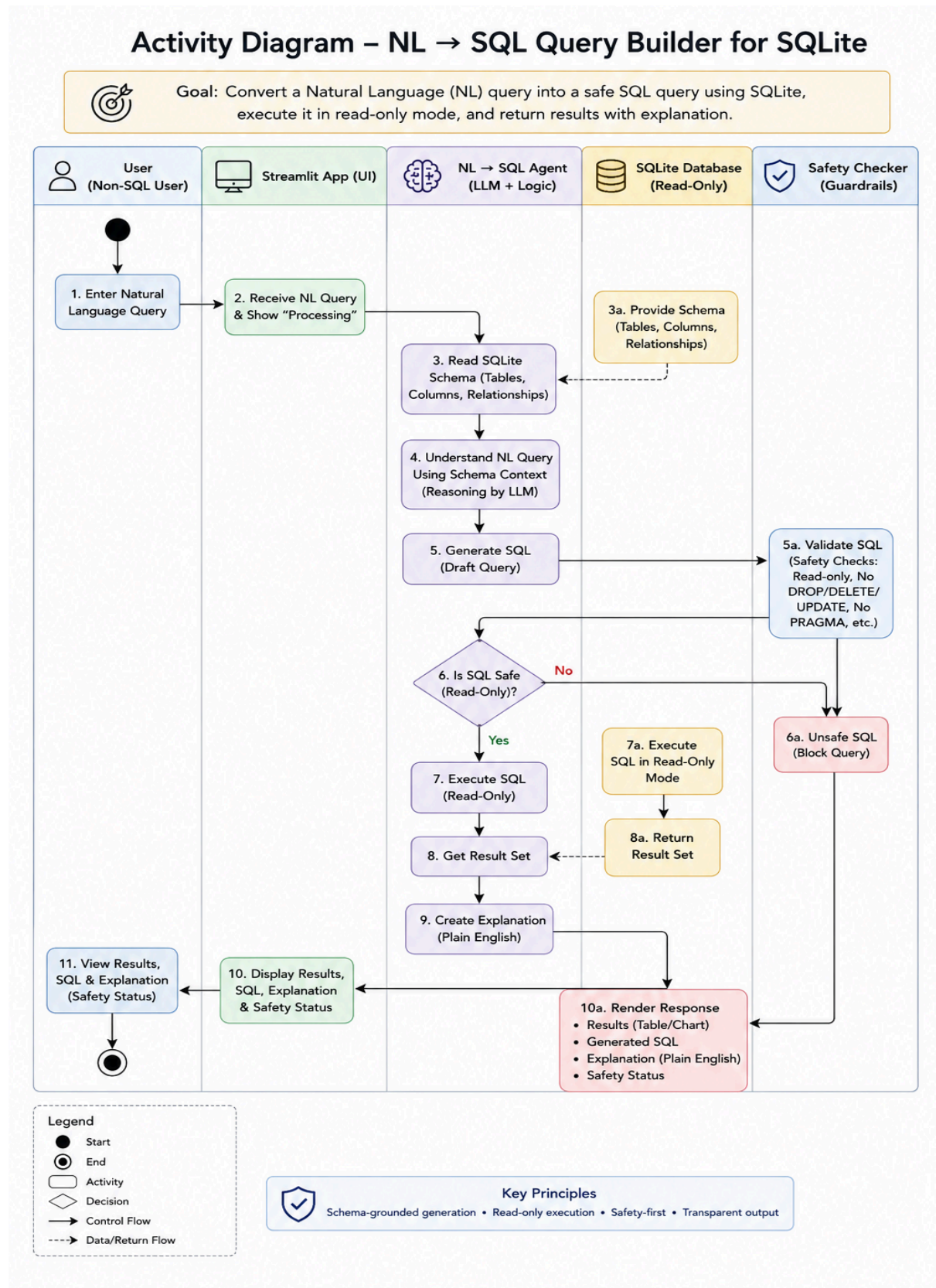
Use Case Diagram:



Sequence Diagram:



Activity Diagram:



Tech Stack Summary

Component	Technology
Frontend	Streamlit
Database	SQLite
Local LLM Runner	Ollama
Agent Framework	LangChain
Data Processing	Pandas
Language	Python

Security & Safety Measures

- Blocks destructive commands such as:
 - DROP
 - DELETE
 - TRUNCATE
 - ALTER

- UPDATE
- INSERT
- Allows only read-only operations.
- Prevents accidental database modifications.
- Validates generated SQL before execution.

Challenges Faced

- Accurate interpretation of natural language queries.
- Handling ambiguous user requests.
- Maintaining SQL generation accuracy.
- Optimizing response time for large databases.
- Ensuring database security during execution.

Outcomes & Benefits

- Reduces dependency on SQL expertise.
- Improves productivity for non-technical users.
- Enables faster data retrieval.
- Enhances decision-making through easy access to information.

- Provides transparency by displaying generated SQL.

Conclusion

- The AI SQL Copilot simplifies database interaction by converting natural language into SQL queries.

It enables users without SQL expertise to retrieve data efficiently.

The system ensures safe query execution through validation and security checks.

It improves productivity, accuracy, and accessibility of database operations.

The project demonstrates the practical application of AI and NLP in data management.

Overall, it provides a user-friendly and intelligent approach to database querying.